



histopathology introduction to pathology

3rd class Lecture 1

Assistant Prof. Dr. Rasha A. AL Safi

histopathologist

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Objectives

By the end of this lecture, students will be able to

1. Define pathology, explain its goals, and differentiate between various branches of pathology with accurate examples.
2. Demonstrate the ability to match appropriate pathological techniques (e.g., light microscopy, immunohistochemistry) with relevant diagnostic applications during lab-based scenarios.

Definition : term *pathology* literally is translated as

➤ Logos: is the study of disease.

pathos: suffering.

- The **pathology** of any disease:

1. **Structural changes** of the relevant tissues.
2. Reflected as **functional disturbances**,
3. Perceived as **clinical features**.

❖ The **goal of pathology**:

❖ the identification of the **cause/es** of a given disease (**etiology**), the mechanisms of the disease evolution (**pathogenesis**) that can help in disease ***prevention &/or therapy***.

❖ The **clinical medicine & clinical pathology** are complementary and can not be separated

FIELDS OF PATHOLOGY

1. **Experimental pathology:**
2. **Clinical pathology:** the *approach to the patient's illness* clinically is based on the following sequence of steps:
 - A. Patient's history.
 - B. Examination.
 - C. Investigations .
 - D. Diagnosis .
 - E. Treatment.

Subdivisions of clinical pathology

1. **Histopathology**: concerned with the *investigation* and *diagnosis* of disease from examination of *tissues*.
2. **Cytopathology**; concerned with the *investigation* and *diagnosis of disease* from the *examination of isolated cells*. Generally divided into two main categories:
 - i. Exfoliated cytopathology which deals with *spontaneous cell shedding* e.g sputum (cancer Tb...), urine.
 - ii. Fine needle aspiration in which *cells are extracted* by the aid of *negative pressure aspiration* e.g (breast, thyroid, salivary glands).
3. **Hematopathology**: concerned with the study of disorders affecting the *cells* and the *coagulation system* of *blood*.



5. **Microbiology**; concerned with the study of *infectious diseases* and the *organisms* responsible for them.
6. **Immunopathology**; concerned with the study of *disturbances* affecting the *defense mechanisms* of the body, and their contribution to the disease processes.
7. **Chemical pathology**; concerned with the study and diagnosis of disease from the *chemical changes* that occur in tissues and fluids.
8. **Medical genetics**; concerned with the study of *abnormal chromosomes and genes* and their relevance to disease processes.
9. **Toxicology**; concerned with the study of the effects of known or suspected *poisons on the body*.

10. Forensic pathology: concerned with the application of pathology to ***legal purposes*** (e.g. investigations of **death** in suspicious circumstances).

TECHNIQUES OF PATHOLOGY

1. Gross pathology (macroscopic pathology).
2. Light microscopy.
3. Histochemistry.
4. Immunohistochemistry and immunofluorescence
5. Electron microscopy:
6. Biochemical techniques:
7. Hematological techniques:
8. Cell cultures:
9. Medical microbiology:

1. Gross pathology (macroscopic pathology):

- The changes affecting various organs and tissues in diseases as evident to the ***naked eye***.
- The gross pathology of many diseases is so characteristic that an experienced pathologist can give a fairly confident diagnosis of the disease before further investigations are carried out.

2. Light microscopy:

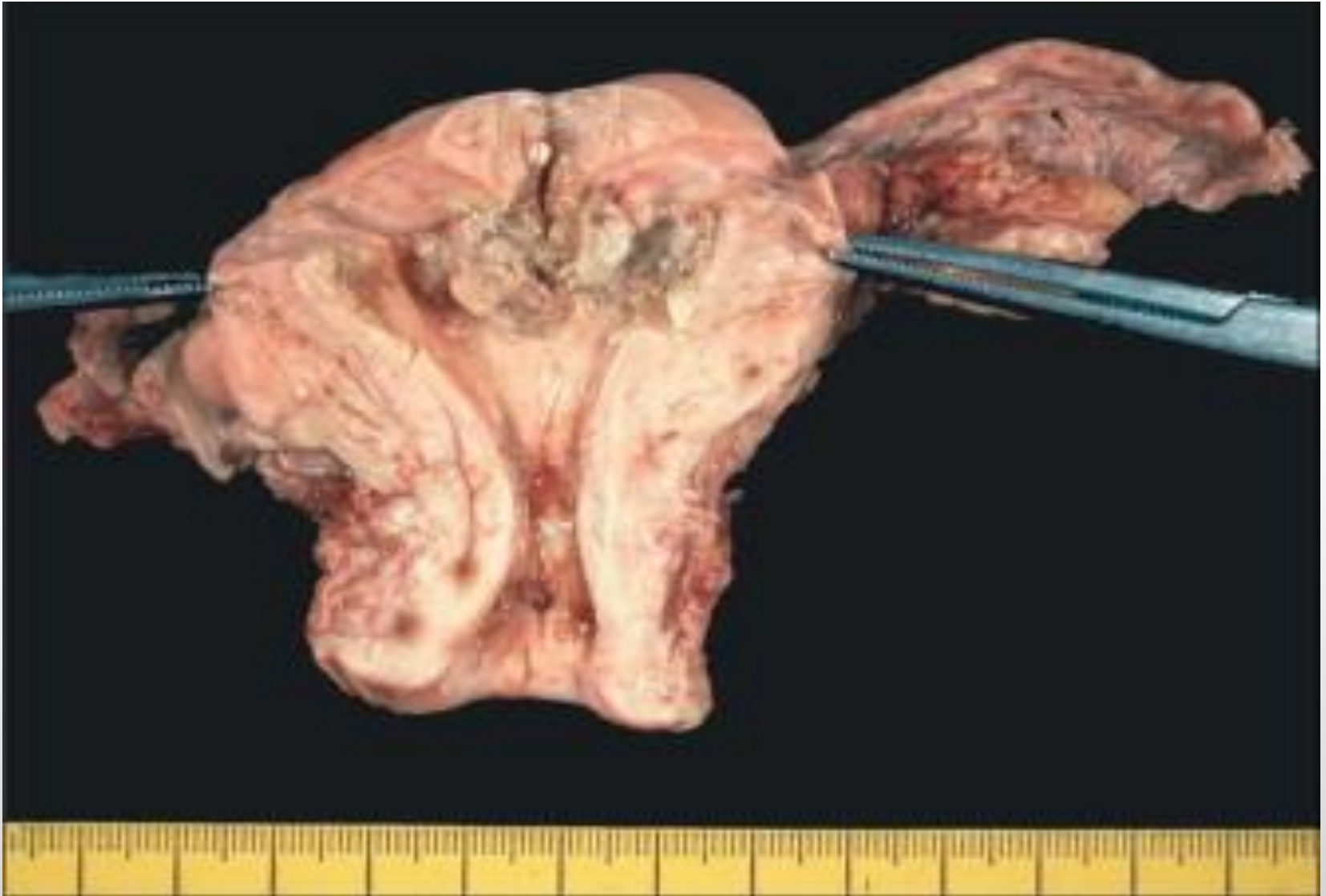
- Use the light microscope in the examination ***tissue structure*** and ***cells*** in health and disease.

The gross appearance of
lung:

- A patient with organizing **diffuse alveolar damage**.
- known as "**honeycomb**" **lung** because of the appearance of the **irregular air spaces** between bands of **dense fibrous tissue**.



Gross appearance of **endometrial carcinoma**



Microscope Parts

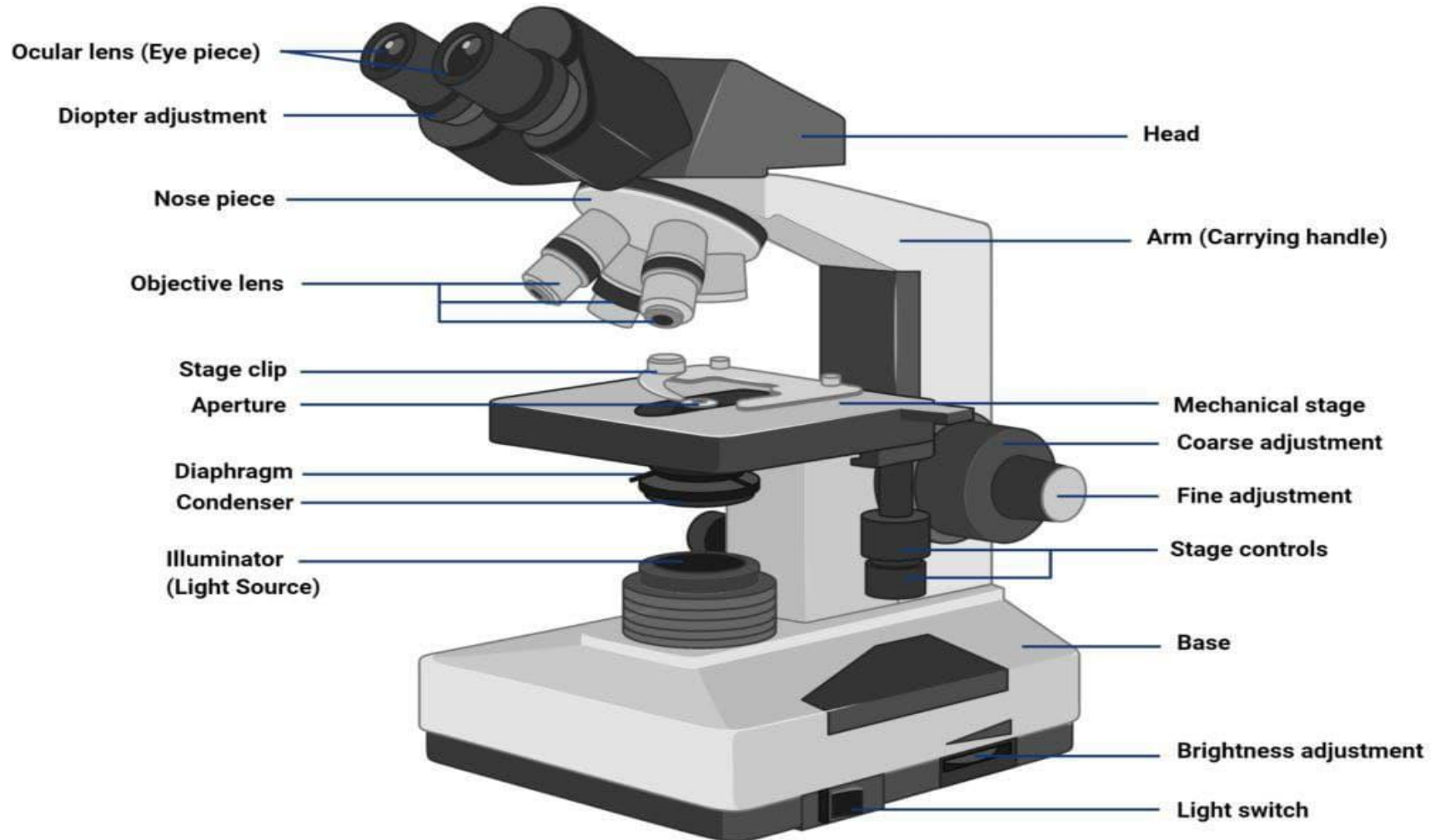


Figure: Parts of a microscope, Image Copyright © Sagar Aryal, www.microbenotes.com

Specimen preparation for histopathological examination

1. Tissue sectioning



2. Tissue imbedding

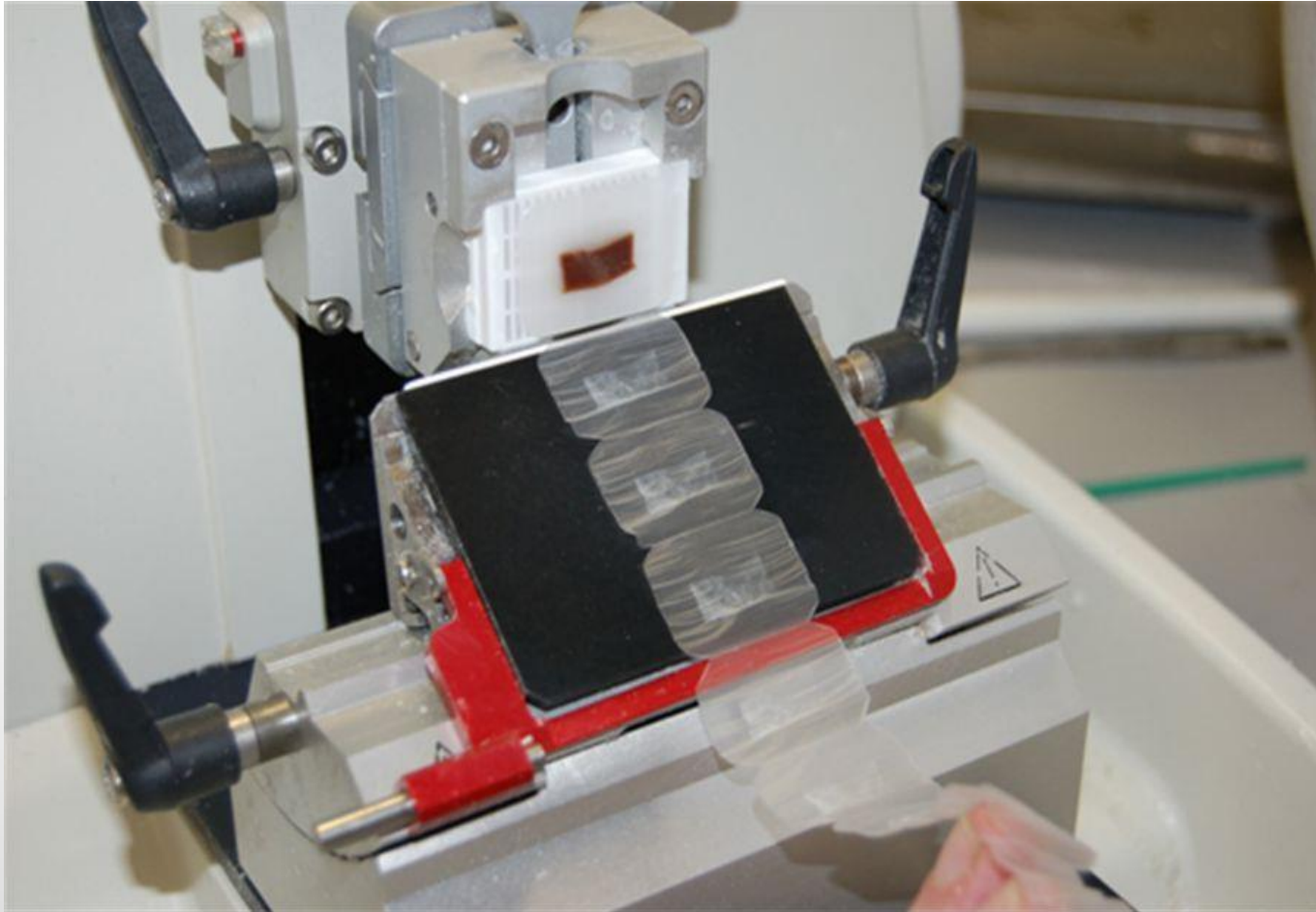


- If **solid tissues** (e.g. liver, kidney etc.) are to be examined by light microscopy, the sample must first be:
 1. **Embedding in wax**: to facilitate hardening and cutting sections from tissue to be examined.
 2. **Thinly sectioned** to:
 - a. Permit the *transmission of light* and
 - b. Minimize the *superimposition of tissue components*.

3. Histochemistry:

- certain cells produce **chemical substances** that detected by specific reagents (**histochemical stains**) is of diagnostic value.
- E.g. **PAS stain** (for **glycogen**).
- **Perl's stain** (for **iron**).

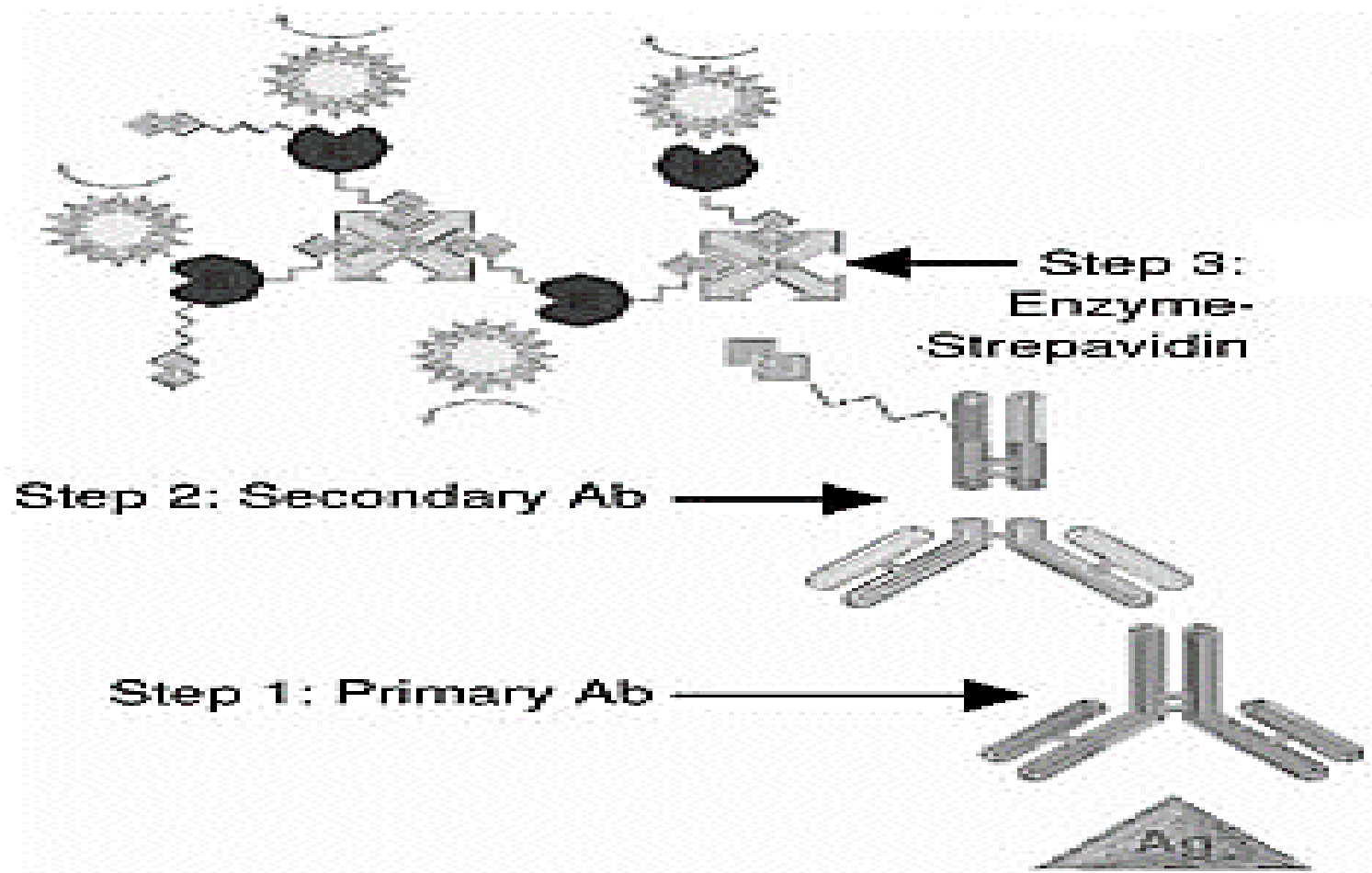
Sectioning by microtome



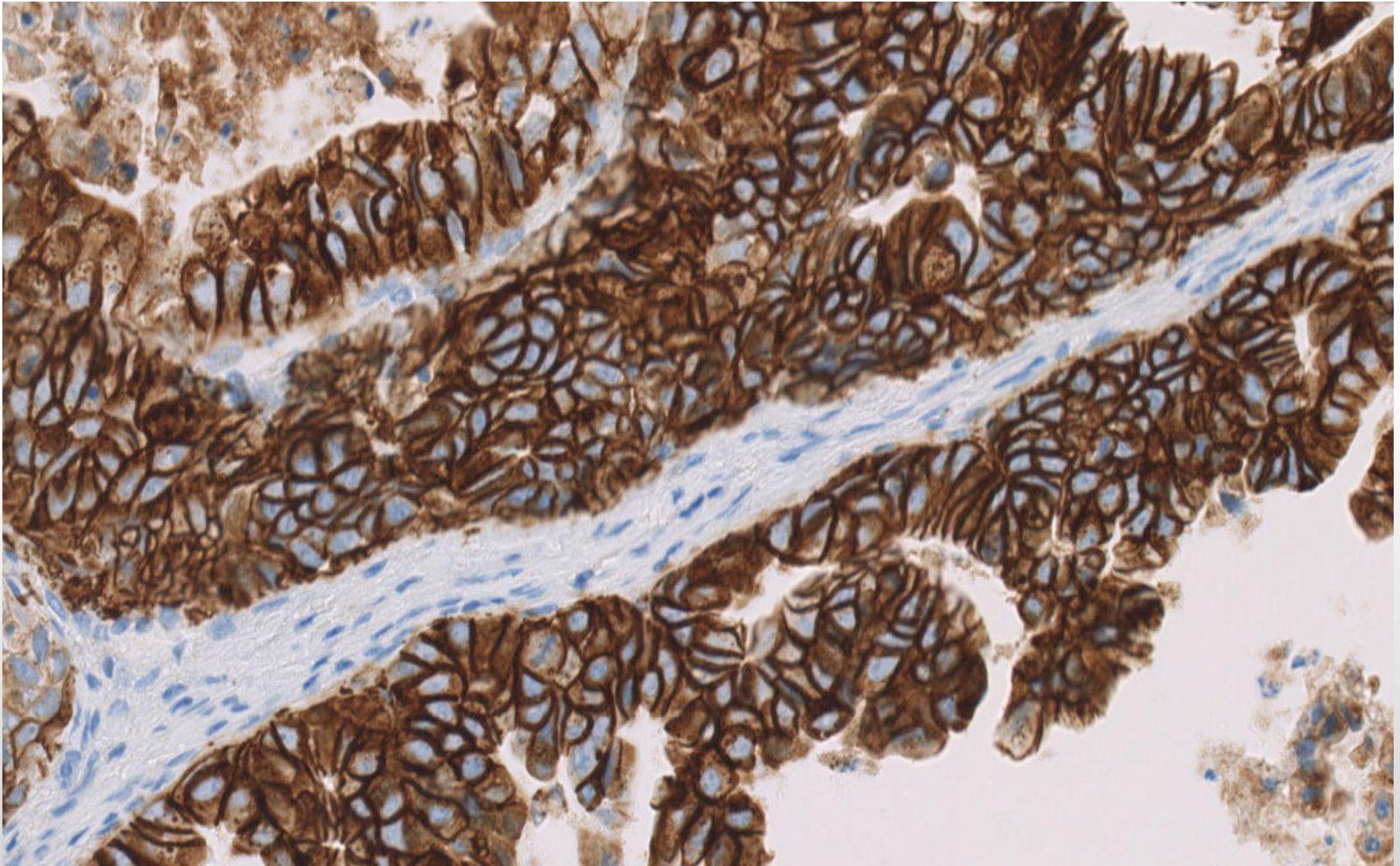
4. Immunohistochemistry and immunofluorescence

- These techniques use **antibodies** (Ag specific) and connected chemically to **enzymes** to **visualize** substances (for e.g. **cellular proteins** or **surface receptors**) in tissue sections or cytological cell preparations.
- The end product is a **deposit of colored material** that can be seen with a conventional light microscope.
- When **antibodies** are connected chemically to **fluorescent dyes** its called (**Immunofluorescence**) technique.
 - The list of substances detectable by these techniques has been greatly enlarged by the development of **monoclonal antibodies**.

Principle of IHC



Her2/neu protein expression in breast cancer (membrane staining)



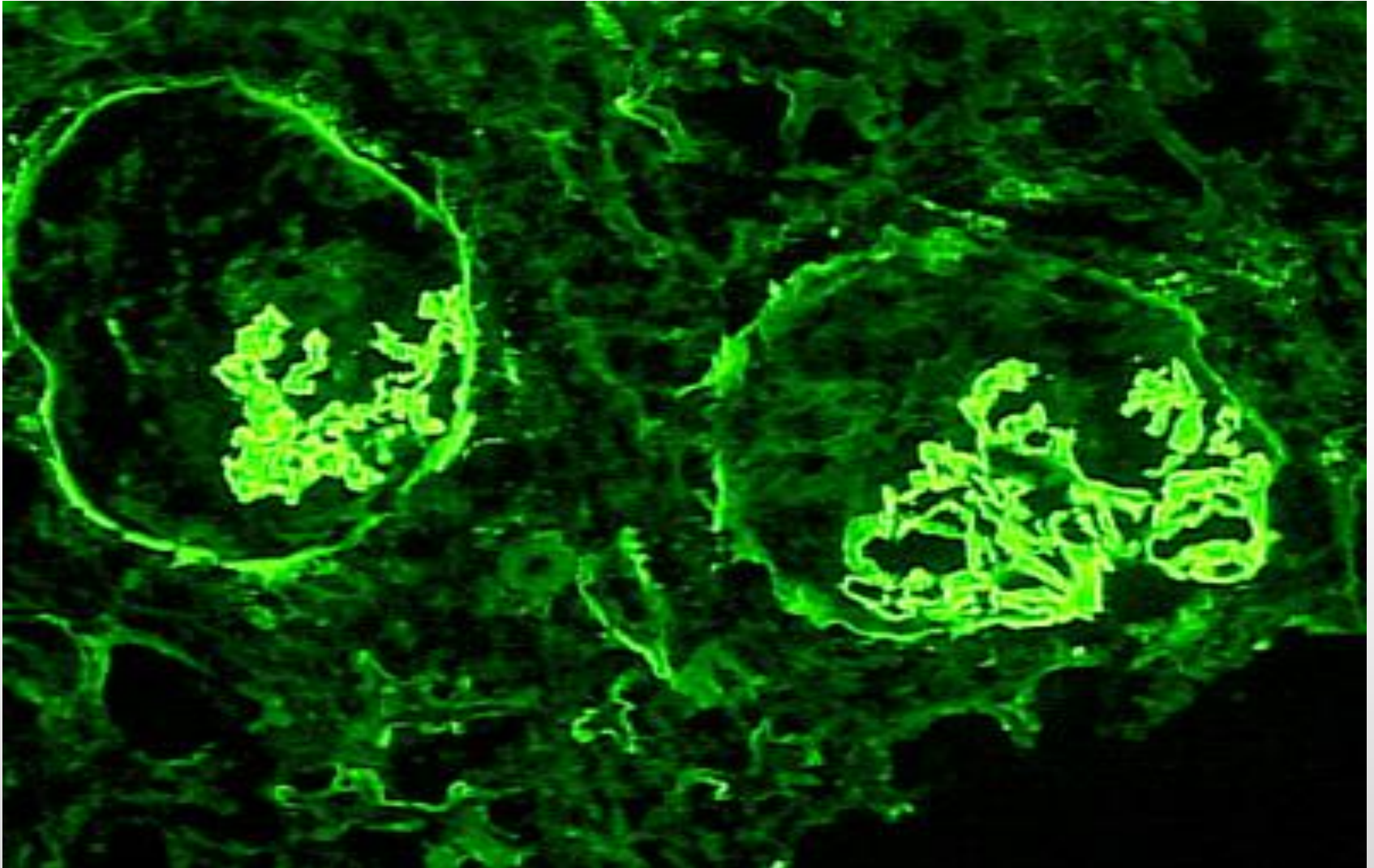
5. Electron microscopy:

- Used to study disorders at an **organelle** (subcellular) level and the demonstration of **viruses in tissue** samples from some diseases.
- The most common diagnostic use is the interpretation of **renal biopsies** in (*glomerulonephritis*).

6. Biochemical techniques:

- These are applied to the **tissues** and **body fluids**.
- The **clinical role** of biochemistry:
 1. Monitoring **fluid** and **electrolyte**.
 2. Detect **serum enzyme changes**: e.g raised levels of cardiac enzymes in the blood indicate damage to cardiac myocytes; the diagnosis of **myocardial infarction (MI)**.

Glomerulonephritis



7. Hematological techniques:

- these are used in the diagnosis and study of **blood disorders**.
- They range from relatively simple **blood cells counting**, to the more sophisticated assays of blood **coagulation factors**.

8. Cell cultures:

- ✓ They are an attractive **medium for research** because of the ease with which the cellular environment can be modified and the responses to it monitored.
- ✓ **Diagnostically**, cell cultures are used to prepare **chromosome spreads** for cytogenetic analysis.

9. Medical microbiology:

- ✓ Study of diseases caused by **organisms** such as **bacteria, fungi, viruses** and **parasites**.

Molecular pathology:

- Advances in molecular pathology is helpful for e.g. to clarify that the defects in the **chemical structure** of **molecules** are the result of **errors** in the **genomic DNA**.

1. In situ hybridization technique (ISH):

- It is possible to make the presence of **specific genes (on DNA)** or their **mRNA visible** in tissue sections or cell preparations.

2. DNA microarrays:

- Can be used to determine ***patterns of many gene expression at the same time*** (mRNA expressed by genes).

3. Polymerase chain reaction (PCR):

- Minute** quantities of **nucleic acids** can be **amplified** by the use of (PCR).

Building knowledge and understanding

- A **logical** and **orderly way of thinking** about diseases and their characteristics must be depended.
- For each disease entity the student should be able to list the chief characteristics:
 1. Epidemiology
 2. Etiology
 3. Pathogenesis
 4. Pathological and clinical features
 5. Complications and sequelae
 6. Prognosis

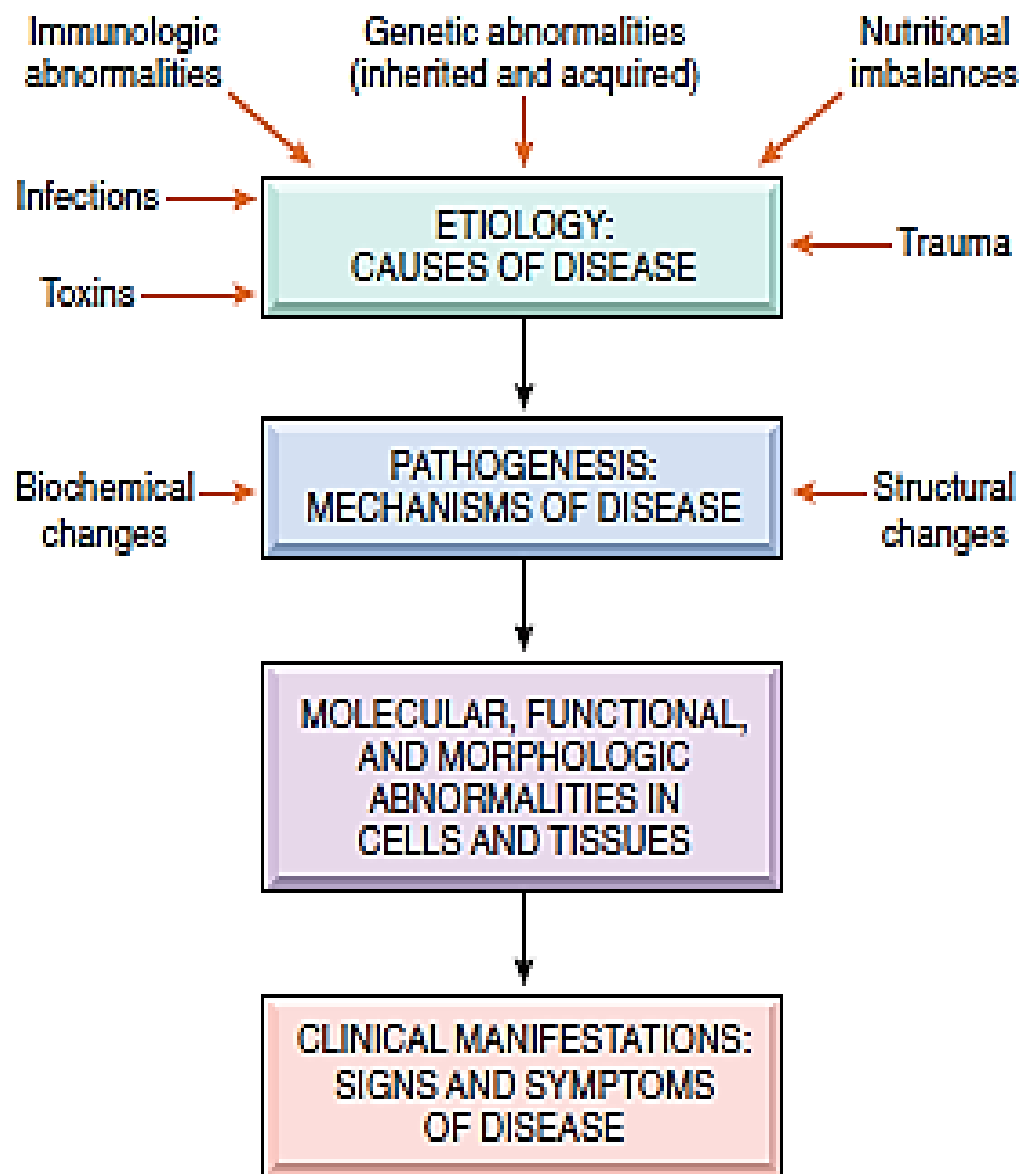


Fig. 2.1 Steps in the evolution of disease. Only selected major causes (etiologies) are shown.

